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Article in Water Research · April 2024
DOI: 10.1016/j.watres.2024.121572

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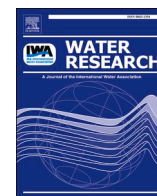


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Spread of antibiotic resistance genes in drinking water reservoirs: Insights from a deep metagenomic study using a curated database

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ARTICLE INFO

Keywords:

Antibiotic resistant gene
Drinking water reservoir
ARG carriers
Metagenomic assembled genome

ABSTRACT

The exploration of antibiotic resistance genes (ARGs) in drinking water reservoirs is an emerging field. Using a curated database, we enhanced the ARG detection and conducted a comprehensive analysis using 2.2 Tb of deep metagenomic sequencing data to determine the distribution of ARGs across 16 drinking water reservoirs and associated environments. Our findings reveal a greater diversity of ARGs in sediments than in water, underscoring the importance of extensive background surveys. Crucial ARG carriers—specifically *Acinetobacter*, *Pseudomonas*, and *Mycobacterium* were identified in drinking water reservoirs. Extensive analysis of the data uncovered a considerable concern for drinking water safety, particularly in regions reliant on river sources. Mobile genetic elements have been found to contribute markedly to the propagation of ARGs. The results of this research suggest that the establishment of drinking water reservoirs for supplying raw water may be an effective strategy for alleviating the spread of water-mediated ARGs.

1. Introduction

The extensive and often indiscriminate usage of antibiotics has led to the emergence of antibiotic-resistant bacteria, which pose a major global human health threat. This problem has become especially acute during the COVID-19 pandemic, because antibiotics have frequently been administered even without there being clear clinical indications (Strathdee et al., 2020). Antibiotic resistance genes (ARGs) are now commonly detected in diverse environments, particularly in water-based ecosystems affected by human activities (Li et al., 2017; Paulus et al., 2019; Reddy and Dubey, 2019; Wang et al., 2019; Yang et al., 2021, 2020). Horizontal gene transfer (HGT) is a major factor in

the spread of these ARGs (Fan et al., 2020; Karkman et al., 2018; Kück et al., 2014). Additionally, the safety of potable water for a substantial portion of the global population is precarious because water is being contaminated by ARG-carrying pathogens. Developing countries may encounter an increased susceptibility to the proliferation of ARGs, particularly in Asian countries characterized by high population densities (Guo et al., 2023). This contamination not only complicates the treatment of infections but also contributes to the emergence of super-bacteria. Alarming projections suggest that by 2050, bacterial infections with ARGs could become the leading causes of mortality (Humphreys and Fleck, 2016). To mitigate the hazards posed by multidrug-resistant pathogens, researchers have advocated for

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legislative measures and the promotion of in addition to promoting responsible antibiotic use (Ayukekbong et al., 2017; Mo et al., 2017). Despite these efforts, ARG contamination of groundwater and rivers has been widespread due to large-scale wastewater discharge and insufficient regulation (Xu et al., 2019). In China, drinking water reservoirs are typically constructed along local rivers and natural lakes, constituting a subset of water supply reservoirs characterized by more stringent environmental requirements. Divergent from finished-water reservoirs, drinking water reservoirs primarily furnish untreated water to drinking water plants. They are distinguished by a water collection inlet directly connected to drinking water plants and a drainage outlet directed downstream. This alignment with the normative definition of drinking water reservoirs as articulated by Feldbauer and colleagues (Feldbauer et al., 2020). Notably, these reservoirs serve as a vital source of potable water for a majority of Chinese urban residents. Owing to regulatory policies and considerations related to public safety, the collection of samples from drinking water reservoirs is often restricted, particularly within 500 m from the raw water collection inlet. Comprehensive sampling is crucial to understanding the extent of ARG contamination and to develop effective strategies, including the establishment of a legislative framework, for combating the escalating ARG threat. In our big data analysis and other investigations pertaining to drinking water reservoirs, the insufficient sample sizes in previous studies have not fully elucidated the spectrum of ARGs distribution and their environmental driven factors (Guo et al., 2023). On the contrary, the current discovery/identification and subsequent update of ARG databases help us to close the gap in ARG studies.

Most large reservoirs (the capacity > 100 million m³) are situated on main river channels and serve purposes such as hydropower generation, irrigation, and flood control. By contrast, drinking water reservoirs are established according to local water systems, including tributaries or natural lakes. Since the 2010s, Zhejiang Province has systematically performed measures for protected areas surrounding drinking water reservoirs. According to the protection level, these measures may contain the installation of barbed wire, restrictions on animal husbandry, discouragement of settlements by non-residents, and the centralized collection and treatment of domestic and medical wastewater from villages. The abundance and diversity of ARGs are often higher in reservoirs than in their input streams, largely due to a lower flow rates and longer retention times in reservoirs (Huang et al., 2019b). Raw water is conveyed to drinking water treatment plants via water inlet pipelines, thereby furnishing potable water to many people in China. ARGs carriers in sediments exhibit a tendency for long-term survival, influencing the ARGs distribution in water (Deng et al., 2020). Presence of antibiotic contamination may further facilitate to occupy ecological niches by corresponding ARG carriers. Current water purification and treatment measures are not specifically designed to eliminate ARGs and may accidentally enrich the ARG carriers and contribute to their mutation (Sanganyado and Gwenzi, 2019; Wan et al., 2022). The raw water influences the quality of drinking water, and the ARGs in drinking water can undergo HGT into gut microorganisms (Khan et al., 2020). Even the resistome in drinking water often exhibit similarities to resistance of infectious pathogens clinically identified in local hospitals (Dias et al., 2020; Diorio-Toth et al., 2023). Metagenomic sequencing offers unique advantages in identifying uncultured host organisms, analyzing evolutionary patterns, and understanding HGT mechanisms in microbial communities (Dang et al., 2020). Hence, it is imperative to elucidate the profiles, transfer mechanisms, and carriers of ARGs in drinking water reservoirs. Such insights are crucial for mitigating human exposure to water-mediated ARGs, reducing mortality associated with multidrug-resistant pathogens.

Although researchers have undertaken some exploratory efforts (Dang et al., 2020; Saleem et al., 2018), their endeavors remain inadequate due to the ecological diversity inherent in drinking water reservoirs and the significance of the issue (Guo et al., 2023). We conducted a comprehensive sampling campaign, focusing on sediments and water

from 16 reservoirs near drinking water inlets across Zhejiang Province, China (Figure S1). This study significantly contributes to the current understanding of ARGs in drinking water reservoirs through a multifaceted approach encompassing the following key elements: (1) development of an enhanced curated database tool for improved ARG detection and comprehensive identification of mobile genetic elements (MGEs), with the database now available to the public; (2) employment of deep-metagenomic sequencing (totaling 2.2 Tb across 36 samples) to establish a detailed profile of ARG in these drinking water reservoirs; (3) conduction of large-scale data analysis of ARGs, involving 50 groups of metagenomic sequencing data collected from global drinking water reservoirs and 600 groups of data from 12 environmental conditions associated with drinking water; and (4) investigation of potential pivotal ARG carriers, their habitat preferences, and driving force. Overall, this study has enhanced the detectability and established a new framework for analyzing ARGs in drinking water reservoirs. Our findings illuminate that river water may not serve as an optimal raw water source for the prevention of ARGs spread and multidrug-resistant bacteria transmission, also highlights the establishing drinking water reservoirs in ensuring the safety of drinking water for residents. The insights can inform the development of strategies and legislation aimed at mitigating the effects of antibiotic resistance on public health.

2. Results

2.1. ARGs profiles across drinking water reservoirs

To enhance the identification of ARGs, we have introduced a novel open-source tool named ARGs-MGEs-MAP (AMM), effectively doubling the count compared to the SARG database v2.2 (Yin et al., 2022) while maintaining similar levels of Matthews Correlation Coefficient (MCC; 0.9345), precision (0.9679), and sensitivity (0.9028; Fig. 1a-c). A comprehensive description of AMM and its comparison to SARG is provided in the supplementary material (Supplementary text and Figure S2). In this study, we utilized the curated database AMM, and extensive sequencing data totaling 2.2 Tb to provide a detailed overview of the ARGs and MGEs across 16 drinking water reservoirs in Zhejiang Province, China.

We identified 17 distinct types of ARGs in all analyzed samples (Fig. 1d), with abundances ranging from 2.89×10^{-6} to 0.467 copies per cell. In both sediment and water samples, the levels of genes conferring resistance to multiple drugs and bacitracin were consistently highest among all gene levels. The sediment samples had a greater diversity of ARG subtypes than the water samples did (Fig. 1e-f). Specifically, we observed a broader resistome within the sediments, and a clear difference in ARG subtypes and their co-occurrence patterns between water and sediment habitats, along with a higher rate of positive correlations among ARGs in sediment samples (78.7 %) compared to water samples (42.9 %; Figures S3 and S4). These findings suggest that sediments are likely to serve as endogenous sources of ARGs. Compared to related conditions, the abundance of ARGs in drinking water reservoirs was found to be lower than that in most other environmental conditions (Fig. 2a). ARGs were found to be significantly more abundant in wastewater, whereas river water and drinking water had more than double the ARG abundance of that in drinking water reservoirs.

2.2. Investigation crucial ARG carriers and potential driving forces of ARG spread

To understand the factors influencing the distribution of ARGs, we conducted the principal coordinate analysis (PCoA) and the Procrustes transformations. The PCoA revealed considerable differences between the water and sediment samples in terms of ARGs, MGEs, and microbial taxa (Figure S5a-c). Procrustes transformations based on the Bray-Curtis distances in PCoA matrices, revealed correlations among ARGs, MGEs, and taxa (Fig. 2b and S5d, e). The strongest correlation was that between

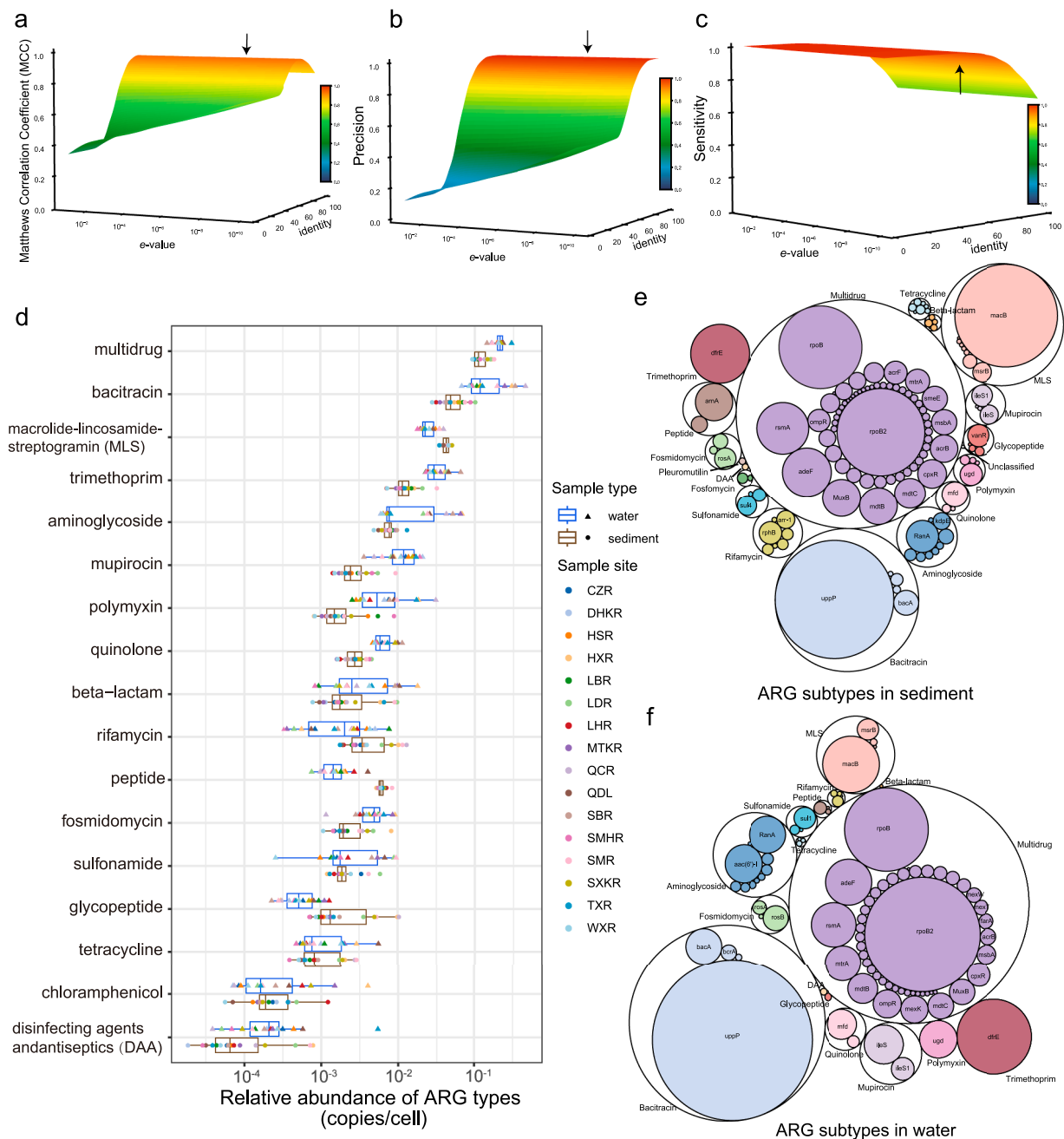


Fig. 1. (a-c) Assessment of the ARG annotation and quantification using the curated database AMM, evaluating (a) Matthews Correlation Coefficient (MCC), (b) sensitivity, and (c) precision with a 150 bp read length. The color gradients represent the simulated values, and the cutoff in use is indicated by black arrows. (d-f) Depicting the ARG profiles in the drinking water reservoir samples. (d) Boxplots display the relative abundances of ARG types in reservoirs, while separately displaying the water and sediment samples (copies/cell), respectively (water: triangle spots and blue boxes; sediments: round spots and brown boxes). (e-f) Composition and distribution of ARGs in (e) sediment samples and (f) water samples. The ARG composition is depicted by color according to their respective types. The outer and inner circles represent ARG types and subtypes, respectively, with the circle size being proportional to the normalized abundance of ARGs. The MLS refers to macrolide-lincosamide-streptogramin, and the DAA refers to disinfecting agents and antiseptics.

ARGs and MGEs (having the lowest Procrustes squared residual), and this result prompted a detailed analysis of their quantities based on metagenomic sequencing data (Fig. 2c). This analysis revealed a positive correlation ($r = 0.757$) between the abundances of ARGs and MGEs, despite varied sample backgrounds. Regarding drinking water reservoirs, a significantly positive correlations between ARGs and MGEs was confirmed in both sediment and water ($r = 0.954$ and $r = 0.968$, respectively; Figure S6), highlighting the pivotal role of MGEs in the distribution of ARGs. We elucidate the impact of eight potential variables (water temperature, inlet depth, dam height, reservoir capacity,

catchment area, MGE abundance, latitude, and longitude, showed in Table S1) on the abundance of ARGs (Figure S7). Among these variables, only MGE abundance exhibits a significant effect on ARG abundance in both water and sediment, along with a linear positive correlation (Figure S7a-c). While dam height may potentially influence ARG abundance in sediments, a clear correlation is not readily discernible (Figure S7d). Analysis of Similarities (ANOSIM) highlights the considerable influence of sample types on ARG abundance (Figure S7e).

To further elucidate the distribution and prevalence of ARG carriers, we analyzed 21,036,726 contigs from drinking water reservoirs and

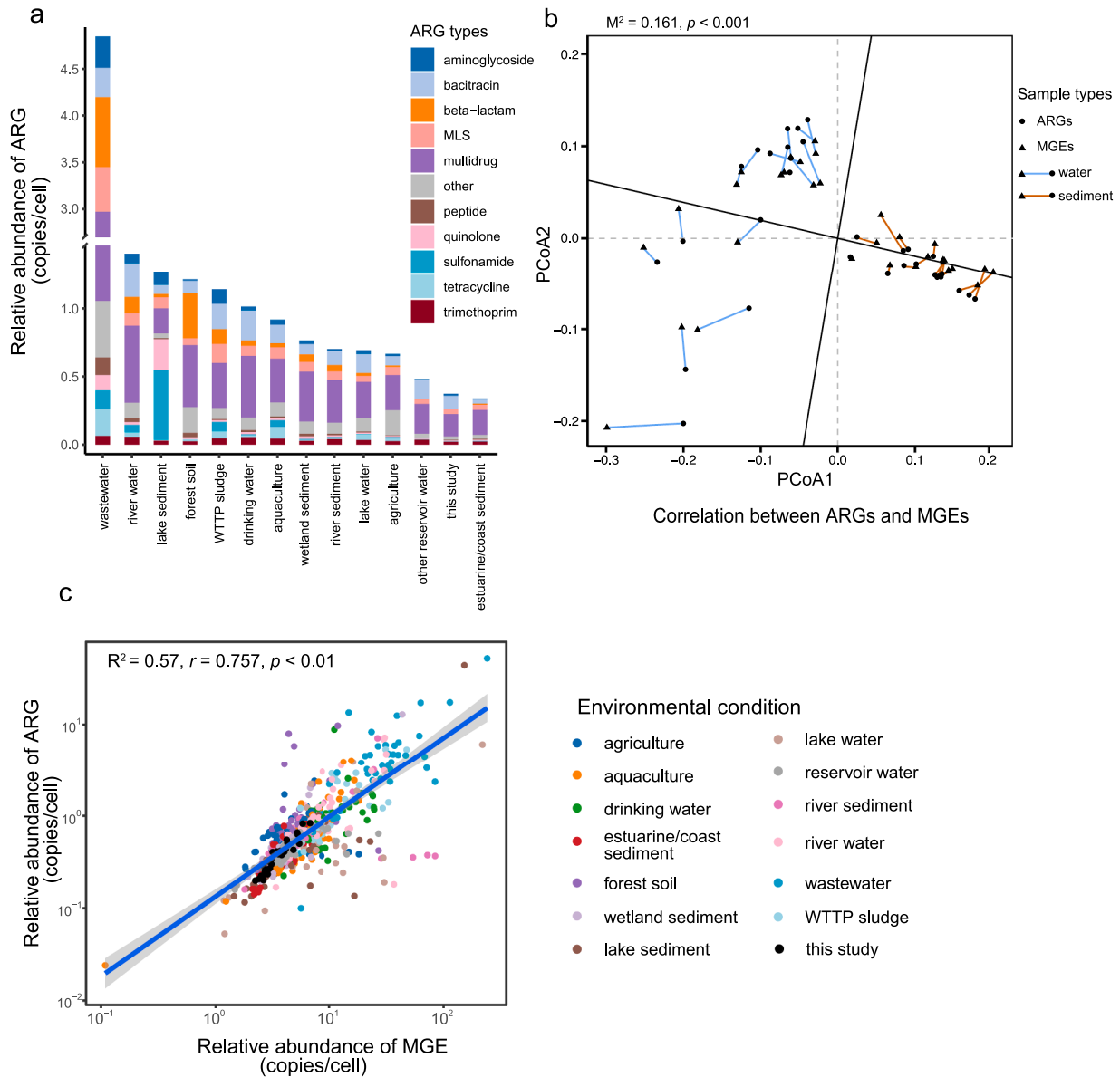


Fig. 2. (a) The evaluated average relative abundance of ARGs and the proportion of ARG types across environmental conditions associated with drinking water. (b) The Procrustes analysis indicates the correlations between the involved ARGs and MGEs with Bray-Curtis distances ($M^2 = 0.161, p = 0.001$). The shape of the spots indicates the sample type (water: triangle spots; sediment: round spots), and the spots are depicted by color according to their sampling sites. (c) The quantitative relation of collected reads between ARGs and MGEs in relative abundance between ARGs and MGEs within environmental conditions relevant to drinking water.

96,299,110 contigs from associated environmental conditions (Table S2). The ARG-carrying contigs from drinking water reservoirs are depicted in Fig. 3a. Notably, ARGs such as *aac(6')-I*, *bacA*, *adeF*, *rpoB2*, *uppP*, *udg*, mupirocin, and quinolone exhibited a clear preference for the water habitats, with over 80 % of their ARG-carrying contigs found in the water samples. This aquatic preference—particularly for *rpoB2*, *udg*, and *uppP*—was corroborated in an extensive range of environmental conditions, indicating a clear tendency for their hosts to thrive in aquatic habitats (Figure S8a).

For the most prevalent ARGs in drinking water reservoirs (*uppP* and *rsmA*), we employed the maximum likelihood method to construct phylogenetic trees on the basis of sequence similarities, aiming to investigate the potential of these ARGs for HGT (Fig. 3b-c). Evidence of transferred undecaprenyl pyrophosphate phosphatases (*uppP*) was detected in individual nodes displaying the highest similarity to clades from another phyla or classes (HGT-1 to HGT-4 events indicated by red triangles in Fig. 3b). The analysis also revealed cross-order HGT events

involving *rsmA* (HGT-5 and HGT-6, indicated by red triangles in Fig. 3c). We analyzed genomic regions adjacent to the identified ARGs (*uppP* or *rsmA*) to confirm the possibility of HGT. Our findings lend supported to the hypothesis that HGT serves as a driving force for the dissemination of ARGs. This is indicated by the presence of similar gene clusters and sequences in microorganisms that are distantly related, as opposed to those found in other members of the same strain (Figure S8b-g). Additionally, our analysis revealed that 149 ARGs loci in same contig with MGEs, with over a half proportion of these loci being *uppP* (23.5 %) and *rsmA* (32.9 %; Table S3). To illustrate potential transferred ARGs and their MGE counterparts, we examined the presence of ARGs in same assembled contig co-occurred with MGEs. Specifically, *rsmA*, *uppP*, and *udg* pose a higher risk of transfer in drinking water reservoirs, while *recA* and *int* are the MGEs most likely associated with HGT events (Fig. 4a-b). Across diverse environments, *rsmA* exhibits a consistently high risk of dissemination, followed by *tetC*, *uppP*, and *sul1*, with *recA* being the most susceptible MGE, followed by *tnpA*, *tnpR*, and *int* (Figure S9a-b).

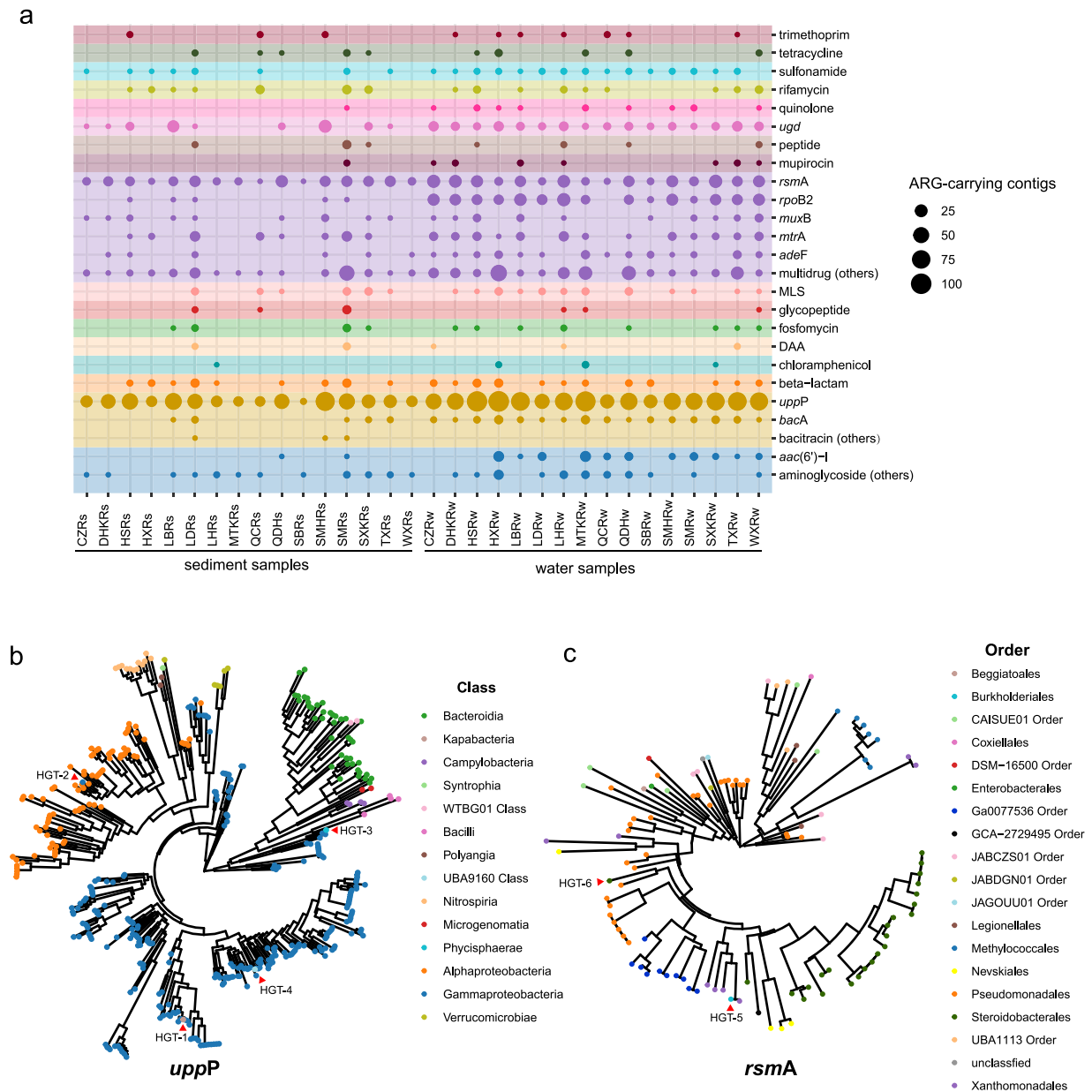


Fig. 3. (a) The distribution of assembled ARG-carrying contigs across drinking water reservoir samples. The size of each spot corresponds to the number of assembled contigs carrying ARGs in the respective sampling sites, with the suffix "-s" denoting sediment samples and "-w" indicating water samples, the color of the spot and underpainting reflects the ARG type. (b-c) Phylogenetic trees were constructed based on the nucleotide sequences of each ARG: (b) *uppP* and (c) *rsmA*; the color of spots represents the taxonomy of the ARG-carrying host (for *uppP*: class-level; for *rsmA*: order-level). The potential HGT events were identified and marked by red triangle symbols (HGT-1 to HGT-6 events).

Our contig-based metagenome assembly process resulted in the generation of 3634 initial bins. These bins subsequently underwent phylum-level taxonomic assignment analysis, which was performed using GTDB-Tk (Parks et al., 2018). Of the metagenome-assembled genomes (MAGs) examined, a significant proportion (40 %, corresponding to 1572 out of 3634) harbored at least one ARG (Table S4). Among these, 1139 bins (31.3 %) satisfied the stringent quality criteria of the minimum information about a metagenome-assembled genome (MIMAG) standard, defined as having a completeness of > 90 % and a contamination level of < 5 %; the phylogenetic tree is presented in Figure S9. Our analysis of ARGs, MGEs, virulence factors, and corresponding phylogenetic relationships of the MAGs carrying ARGs from drinking water reservoir samples (Fig. 4c) revealed three distinctive subgroups: *Acinetobacter*, *Pseudomonas*, and *Mycobacterium*. Despite originating from different sources, these genera tended to cluster

together and were enriched in ARGs, MGEs, and virulence factors (indicated by red lines in Fig. 4c). In drinking water reservoirs, 10 assembled bins, each containing more than 10 ARGs, were exclusively identified as corresponding to *Acinetobacter*, *Pseudomonas*, and *Mycobacterium* (Table S4). According to the MAGs, the most frequently identified ARG types in these genomes included those conferring resistance to multidrug, rifamycin, beta-lactam, fosfomycin, and mupirocin.

2.3. Evaluating the role of drinking water reservoirs in controlling ARGs spread

To assess the effectiveness of drinking water reservoirs in controlling the spread of ARGs, we employed the identical approach to standardize the resistome abundance in each sample across various environmental conditions, including those from drinking water reservoirs. First, using

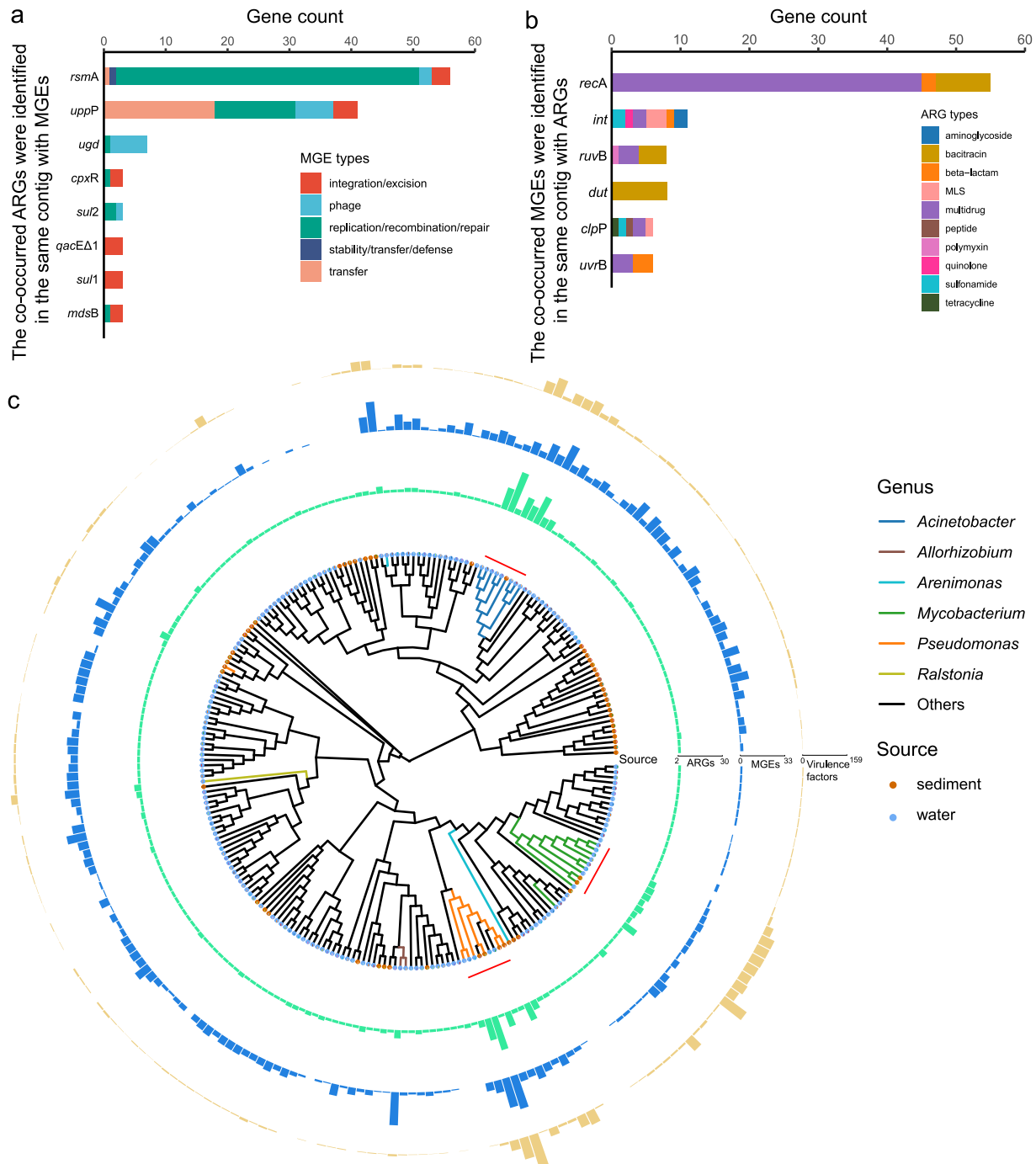


Fig. 4. (a-b) The co-occurred (a) ARGs and (b) MGEs were identified from the same assembled contig in drinking water reservoirs. The color of columns indicates the type of the adjacent genes. (c) The constructed phylogenetic tree was derived from a compilation of 240 assembled bins harboring at least one ARG in drinking water reservoir samples, wherein the colors of connecting lines indicate the genera of hosts. At the terminal points of these lines, the color of nodes symbolizes the source of genomes (water: blue; sediment: brown), and the risky ARG-carrying bins are labeled with red lines. Surrounding the tree's periphery, the outer-circle segments display quantitative representations of harbored ARGs (depicted by green columns), MGEs (depicted by blue columns), and virulence factors (depicted by yellow columns) within each bin.

partitioning around medoids (PAM) for Bray-Curtis dissimilarity clustering, we determined that the silhouette width was greatest when the cluster number (k) was 2, with k increasing up to 600 (Figure S10a-b). The suitability of this binary splitting was further supported by the results of nonmetric multidimensional scaling (NMDS; Fig. 5a-b) and hierarchical clustering (Figure S11), both of which revealed clear separation patterns for ARGs and MGEs. Violin plots (Fig. 5d-e) and heatmaps (Figure S11) indicate significantly greater abundance of ARGs and MGEs in cluster 2 than in cluster 1. Specifically, cluster 2 was found

to reflect significantly higher levels of almost all ARG types (23 out of 26) in abundance and subtype diversity compared with cluster 1 (Figure S12). The higher prevalence and diversity of MGEs in cluster 2 implies greater genome exchange and HGT potential (Figure S13), thereby increasing the likelihood of a superbacterium emerging. Our clustering analysis suggests that the establishment of reservoirs could mitigate the influx of ARGs into drinking water supplies. According to SourceTracker, all samples from drinking water reservoir could be categorized into cluster 1 (Figure S14a-b). Interestingly, cluster 1 and

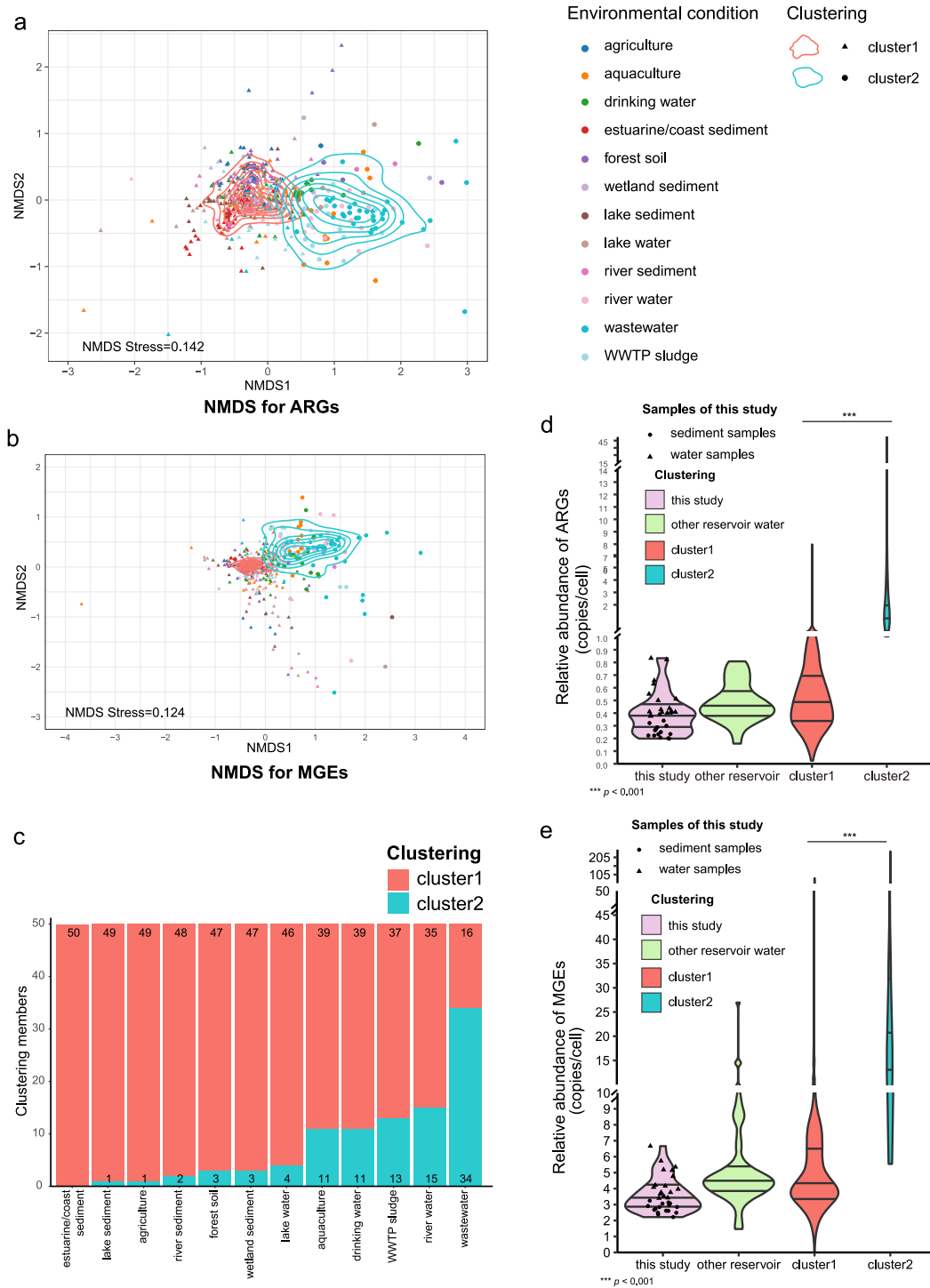


Fig. 5. (a-b) Non-metric multidimensional scaling (NMDS) projections of Bray-Curtis dissimilarities among the (a) ARGs and (b) MGEs across associated environmental conditions. The contours estimate sample densities, and the samples were colored according to their corresponding environmental conditions. The color of the contours represents the cluster assignment based on PAM for Bray-Curtis clustering, with $k = 2$. (c) The stacked bar charts illustrate the occupation of samples across environmental conditions in relation to two identified clusters. The columns are sorted in descending order based on the percentage of cluster1 and the color indicates the corresponding cluster. (d-e) The violin plots exhibit the distribution of relative abundances for (d) ARGs and (e) MGEs among drinking water reservoirs (pink for this study and green for other reservoir water), cluster1 (red violin plots) and cluster2 (blue violin plots). The samples of drinking water reservoirs in this study were denoted by spots (water: triangle spots; sediment: round spots). Statistical significance was determined with the Mann-Whitney U test, with a threshold of $p < 0.001$ indicated by ***.

cluster 2 exhibit distinct variations in the identification of potentially transferred ARGs or potential contributing MGEs through their co-occurrence within the same contig. In cluster 1, *rsmA* remains predominant among ARGs. However, in cluster 2, the abundance of *tetC* and

LamB surpasses that of *rsmA*, and the abundance in cluster 2 exceeds that in cluster 1 by over 10-fold (Fig. 6a). Regarding potential contributing MGEs, cluster 2 shows an over 20-fold abundance greater than cluster1, with *tnpA* and *tnpR* identified as the most active MGEs (Fig. 6b).

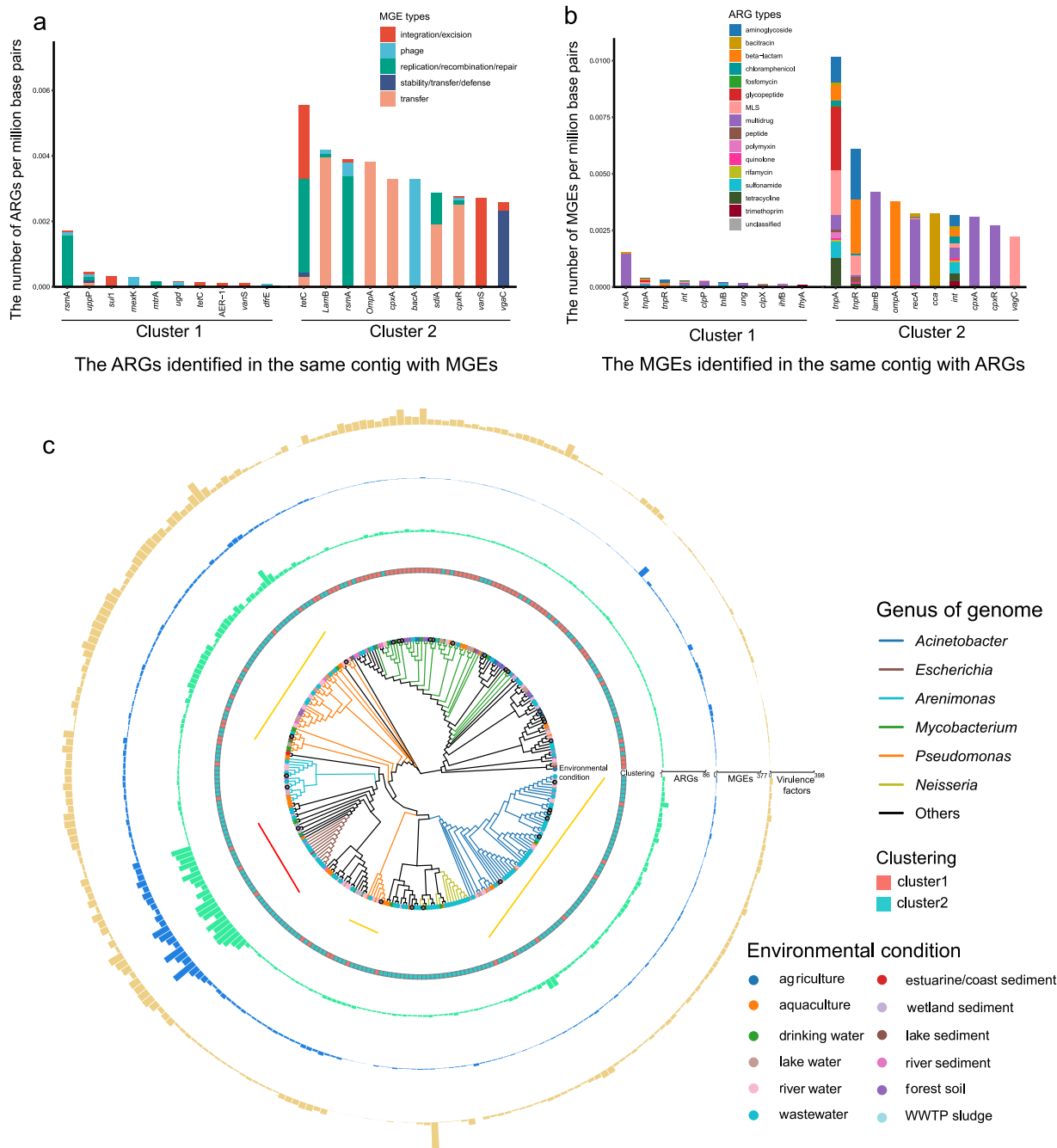


Fig. 6. (a-b) The co-occurred (a) ARGs and (b) MGEs were identified and normalized by million base pairs from the same assembled contig in associated environments. The gene category, indicating whether it belongs to cluster1 or cluster2, is underlined and labeled. The color of columns indicates the type of the adjacent genes. (c) The constructed phylogenetic tree was derived from a compilation of 294 assembled bins harboring at least 5 ARGs from the associated environments, wherein the colors of connecting lines are indicative of the genera of hosts. At the terminal points of these lines, the color of nodes symbolizes the environmental conditions, and the crucial ARG-carrying bins are labeled with red lines. Surrounding the tree, the outer-circle arranged grids display the clustering of these bins (red grids: cluster1; blue grids: cluster2). And the outer-circle columns display the quantitative representations of harbored ARGs (depicted by green columns), MGEs (depicted by blue columns), and virulence factors (depicted by yellow columns) within each bin. The microbiology associated with high ARG risk is delineated by lines (red line: Enterobacteriaceae; yellow line: Pseudomonas and Acinetobacter).

In our investigation into multidrug-resistant bacteria, we analyzed 11,483 MAGs (Figure S14c), of which 3695 were identified as carrying at least one ARG. This analysis supported concerns about the proliferation of superbacteria, particularly the elevated incidence of extreme multidrug-resistant *Enterobacteriaceae* and related species in associated environments (indicated by a red line in Fig. 6c). We discovered 19 cases in which individual genomes contained between 40 and 86 ARGs. Among these, eight originated from wastewater samples and four from

drinking water samples, none of them belong to drinking water reservoir samples. Additionally, the analysis revealed the persistence of multidrug-resistant *Acinetobacter* and *Pseudomonas* in various environmental conditions, within these bacteria containing 5 to 30 ARGs (indicated by the yellow lines in Fig. 6c). On the basis of these findings, we advocate for the consideration of establishing drinking water reservoirs as a strategic measure to reduce the intake of ARGs and prevent the emergence of superbacteria.

3. Discussion

The southern and eastern regions of China are home more than 94 % of the country's population despite constituting only 43 % of its land area (Zheng et al., 2022). This high population density has led to considerable ecological strain and an alarming rise in antibiotic overuse, which presents a formidable challenge to sustainable development. Remarkably, water in Qiantang River and other rivers in Zhejiang Province has been reported to have high concentrations of antibiotics and ARGs (Qiao et al., 2018), further exacerbating the problem of drinking water safety.

In this study, we endeavored to provide a more comprehensive understanding of the types and subtypes of ARGs by using a curated database and applying a specialized pipeline to analyze the ARG profiles of samples from the drinking water reservoirs and associated environments. After a comprehensive analysis (supplementary text and Figure S2), we have doubled the existing capacity for ARG detection without significant compromises in terms of MCC, precision, and sensitivity. Notably, resistance genes for last-line antibiotics such as polymyxin and vancomycin, were found in all samples, despite efforts to since the 2010s to reduce the influx of medical and urban domestic wastewater. The observed differences between sediment and water samples are attributable to various factors, such as fundamental physical and chemical properties (Niño-García et al., 2016), nutrient availability (Chao et al., 2021), biotic interactions (Orland et al., 2019). Because of the higher complexity, our findings suggest that in aquatic ecosystems, sediments function as reservoirs for a diverse array of ARGs (Chen et al., 2019; Dang et al., 2020; Wang et al., 2021). Furthermore, our attribution analysis reveals that the abundance of MGEs and the sample type (water or sediment) exert significant effects on the distribution and abundance of ARGs. Regarding data collection, the acquisition of sufficient metagenomic sequencing data from sediment was limited by the quantity of samples. This limitation highlights that sediment monitoring has been historically overlooked, despite its importance. This neglect and its importance call for immediate attention and integration into standard monitoring practices by drinking water reservoir managers.

Benefiting from the depth of our metagenomic sequencing, we were able to examine the distribution patterns of ARG-carrying hosts distribution patterns and understand their behavior of these hosts in various environments. We identified specific ARG carriers with a preference for inhabiting aquatic ecosystems, especially for the carriers harboring *rpoB2*, *udg*, or *uppP*. This implies that they could be monitored as indicative genes for aquatic antibiotic resistance. We have listed a catalog of ARGs and MGEs with potential transfer contributions, and the findings indicate that *rsmA* and *recA* stand out as the most prevalent ARG and MGE, both in drinking water reservoirs and associated environments. However, in the environments belong to cluster 2 characterized by higher levels of ARGs and MGEs, it is evident that another ARGs and MGEs become more active, with some exhibiting distinct preferences. These high-risk genes (Figs. 4a-b, 6a-b and S9a-b) warrant further investigation by environmental scientists. A key discovery in our research is the strongly positive correlation between ARGs and MGEs, along with evidence of cross-class and cross-order transmission of ARGs. All detected *rsmA* genes belong to the *Gammaproteobacteria* class, which is interesting given the known broader host range of these genes compared to *uppP* (Jeremia et al., 2023). While numerous reports have reported such occurrences, the biological basis of spontaneous HGT among distantly related microbiology remain unclear (Sheinman et al., 2021). The observed spontaneous transfer of ARGs across classes and orders is evident in microorganisms even in environments characterized by the absence of acute antibiotic contamination, low ARG abundance, and the establishment of protected areas. This observations appear to lend support to the hypothesis that "Everything is everywhere but the environment selects" (Baas-Becking, 1934; Shaffer et al., 2022). This suggests that HGT may enable dissemination of ARGs across substantial genetic gaps, even in protected areas, emphasizing the intrinsic role of

HGT as an endogenous driving force in the spread of ARGs.

Our investigation revealed that *Acinetobacter*, *Pseudomonas*, and *Mycobacterium* are crucial ARG carriers in drinking water reservoirs. Their extensive inventories of ARGs and MGEs position them as key nodes in the transfer of ARGs. Such ARG hosts and their pathogenic members are likely to rapidly occupy ecological niches in environments contaminated with antibiotics (Anh et al., 2021). These genera encompass multiple pathogens, including *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Mycobacterium tuberculosis*, which are commonly found in the environment and known to cause infections in humans through direct or indirect contact without proper sterilization (De Jonge et al., 2019; Dijkshoorn et al., 2007; Gygli et al., 2021; Horcajada et al., 2019; Huang et al., 2019a). Given these findings, we propose that *Acinetobacter*, *Pseudomonas*, and *Mycobacterium* be classified as pivotal ARG carriers in drinking water reservoirs and incorporate their antibiotic sensitivity into future environmental risk assessments (Le Page et al., 2017). This categorization is based not only on the frequent presence of ARGs in these organisms but also on the substantial co-occurrence of MGEs within the same genomic contexts, amplifying the role of these organisms as critical nodes in the proliferation of ARGs.

For other associated environmental conditions, we identified numerous instances of extensively multidrug-resistant bacteria in the *Enterobacteriaceae* family (characterized by the presence of > 40 ARGs in their genomes)—such as *Escherichia coli*, *Enterobacter cloacae*, and *Klebsiella pneumoniae*—in habitats including wastewater, river water, and drinking water. This indicates that *Enterobacteriaceae* strains are highly susceptible to ARGs, likely due to the infiltration of drinking water by domestic and medical sewage. Therefore, using river main-streams as sources for drinking water, especially those receiving upstream domestic and medical wastewater, may be inadvisable. Members of *Enterobacteriaceae* family—enriched in ARGs, MGEs, and virulence factors—pose a considerable public health risk, considering their pathogenic ability.

Through our model training and big data analysis, we tried to analysis the effects of anthropogenic activities on the distribution of ARGs and MGEs. The metagenomic data were stratified into two clusters, both ARGs and MGEs exhibited distinct distributions between them, possibly attributed to observed robust correlations. Notably, cluster 2 shows a higher proportion originating from wastewater, river water, wastewater treatment plant (WWTP) sludge, drinking water, and aquaculture, implying the substantial influence of human activities on this cluster. In contrast, cluster 1 exhibits lower ARG abundance, along with fewer members belonging to these environments. In the identification of potential transferred ARGs and contributing MGEs, more than 10 times differences were observed between cluster 1 and cluster 2, along with some genes exhibit clear genetic preferences, warranting further investigation. This, coupled with the identification of extreme ARG carriers, highlights a global concern regarding the safety of current drinking water and its sources. These results reveal the challenges that are being faced by those trying to ensure the safety of human drinking water, particularly in regions reliant on river sources. The upstream contamination of rivers and water resources by animal husbandry, industrial, medical, and urban domestic wastewater is likely contributing to this phenomenon. Given the prevalence of inadequate sanitation and unsafe drinking water in low- and middle-income countries (Sanganyado and Gwenzi, 2019; WHO, 2020), combating the proliferation of ARGs in drinking water is imperative. Drinking water reservoirs, as evidenced by our study and others, exhibit lower risks for ARGs, MGEs, and ARG carriers compared to other associated environmental conditions. Given the reported similarities in resistome profiles between drinking water and those of clinical infectious pathogens (Dias et al., 2020; Diorio-Toth et al., 2023), our study underscores the crucial role of establishing drinking water reservoirs and performing protective measures to mitigate the risk of multidrug-resistant pathogens to humans.

Constrained by regulations, we only examined a singular ARG profile in water and sediment samples from drinking water reservoirs. The lack

of continuous data weakens the comprehensiveness of our perspective. This study relies on the genomic characterization, while the phenotypes exhibited by ARG carriers have not been thoroughly investigated. We also acknowledged that the implementation of protective measures across all drinking water reservoirs may vary from multiple reasons. Thus, conducting comprehensive investigations into ARG backgrounds and tailoring management measures to the local conditions are critical. Considering the limited sequencing data from drinking water reservoirs, we recommend that researchers and monitoring managers upload and share high depth metagenomic sequencing datasets with accurate and detailed labelling include the ARGs and environmental conditions. The lack of datasets remains a bottleneck in current big data research and machine learning. Such efforts could enhance the data infrastructure for the managing of ARGs and public health.

4. Conclusion

We emphasize the potential of sediments as reservoirs for ARG carriers and identify crucial ARG carriers in drinking water reservoirs and associated environments, underscoring the necessity for a comprehensive background survey. Our study highlights a dilemma concerning ARGs in drinking water, where the quality of current drinking water is compromised, and rivers prove to be unreliable sources of raw water. Zhejiang Province provides a notable illustration—the construction of drinking water reservoirs combined with the establishment of protected areas to mitigate human activities, appears to correlate with a reduced abundance and diversity of ARGs. The effectiveness of this reduction is contingent upon the interplay of local environmental conditions and implemented management measures. Therefore, establishing drinking water reservoirs with strict protection measures and the implementation of comprehensive, sustained, and in-depth metagenomic sequencing surveillance is imperative for mitigating the intake of water-mediated ARGs intake.

5. Methods

5.1. Sample collection and DNA extraction

The study was carried out in April and May 2021, and the sampling sites located in the primary protected areas of drinking water source. All reservoirs included in the list of drinking water reservoirs, comprised of Changzhao Reservoir (CZR), Duihekou Reservoir (DHKR), Hengshan Reservoir (HSR), Huaxi Reservoir (HXR), Lubu Reservoir (LBR), Lidun Reservoir (LDR), Lianghui Reservoir (LHR), Matikeng Reservoir (MTRK), Qincun Reservoir (QCR), Qiandaohu Lake (QDL), Shibi Reservoir (SBR), Siminghu Reservoir (SMHR), Shuangmiao Reservoir (SMR), Shuangxikou Reservoir (SXXR), Tingxia Reservoir (TXR), and Wuxi Reservoir (WXR) (Figure S1).

The sampling protocol involved the collecting 10 liters of bottom water and 0.005 m³ of surface sediment from the immediate vicinity of the drinking water inlets, employing cylindrical mud extractor (Juchuang Environmental Protection Group, China) connecting with single piston pump (Henan Yugong Machinery, China). The sterile polymethyl methacrylate containers were then carefully stored under refrigerated conditions (4 °C) to maintain sample integrity for subsequent analyses. Upon collection, the water sample was divided equally into two portions: 5 liters were preserved at 4 °C as a backup, while the remaining 5 liters underwent vacuum filtration utilizing 0.22-μm membranes (Millipore, USA). The filtered membrane water and fresh sediment samples were expeditiously subjected to DNA extraction procedures, while the sediment samples were preserved at 4 °C prior to extraction. The remaining samples were stored at −20 °C to ensure their integrity for subsequent analyses. For DNA extraction, we employed the FastDNA SPIN Kit for soil (MP Biomedicals, USA), following the manufacturer's instructions. The quality of the extracted DNA was assessed using 1 % agarose gel electrophoresis and Nanodrop spectrophotometry

(Thermo Scientific, USA). The DNA extraction from each sample underwent a tripartite process, signifying a technical replication, followed by thorough mixing prior to sequencing.

5.2. DNA sequencing

For the construction of DNA libraries, a quantity of 100 ng of total DNA was utilized, employing the MagicSeq DNA Library Prep Plus Kit (Maibo, China) to generate libraries with a target fragment size of 350 bp. Library concentrations were performed using Magic 1 × ds DNA HS Assay Kit (Maibo, China) After the completion of library quality control, the libraries were pooled based on their effective concentration and the desired amount of targeted data. The 5'-end of each library was subjected to phosphorylation and cyclization, followed by loop amplification to generate DNA nanoballs. These DNA nanoballs were subjected to sequencing using the DNBSEQ-T7 platform with PE150 (paired-end 2 × 150 bp) sequencing strategies, conducted by Hangzhou Woosen Biotechnology Ltd. (Hangzhou, China).

5.3. Publicly available metagenomes and database

To obtain comprehensive ARGs in environmental conditions, a total of 4939 sets of metagenomic samples were collected from the National Center for Biotechnology Information (NCBI) Sequence Read Archive (<https://ncbi.nlm.nih.gov/sra/>). These samples were meticulously categorized into 12 distinct types, namely agriculture, aquaculture, drinking water, estuarine/coast sediment, forest soil, wetland sediment, lake sediment, lake water, river sediment, river water, wastewater, and WWTP sludge. In addition, this study encompassed a set of metagenomic sequencing data derived from water samples of other drinking water reservoirs. To ensure robustness and minimize potential biases, a series of selection procedures were applied, including: (i) an equitable distribution of samples across each type, with 50 samples selected for each category. (ii) Exclusion of samples collected from potentially heavily contaminated environments, such as pharmaceutical manufacturing parks, hospitals, livestock farms, and coal-fired power plants. These environments are known to contain elevated levels of pharmaceuticals, heavy metals, and persistent organic pollutants, which could introduce uncertainty and bias into the datasets by favoring the selection of antimicrobial-resistant strains. (iii) Exclusion of cultured samples or samples subjected to additional experimental processes to maintain the integrity of the natural metagenomic profiles. (iv) Prioritization of samples from different BioProject to ensure representation from diverse sources. (v) Inclusion of only paired-end sequencing reads in FASTQ format for further analysis.

After careful selection, a total of 650 sets of samples were chosen for subsequent analyses, and their detailed information can be found in Table S5. These samples were processed using the previously described methodology outlined in this study. The findings encompassing all metagenome assembly genomes are elucidated within Table S6, the number ARGs, MGEs and virulent factors are list in Table S4.

To assess the influence of sequencing depth, we gathered raw data from 650 metagenomic samples and determined the coverage depth using the median. Then, we randomly selected 12 metagenomic samples from associated environments and used seqkit (<https://github.com/lh3/seqtk>) to randomly extract the corresponding number of reads for analysis.

The curated database tool named ARGs-MGEs-MAP pipeline (AMM) in this study has been published on Github (<https://github.com/bioone/ARGs-MGEs-MAP>).

5.4. Metagenomic sequencing analysis

A total of 2.2 TB of clean sequencing data were obtained from 32 samples, with a basic information and a summary of the datasets listed in Table S1 and S2. To get clean reads, the reads for each sample were

subjected to quality filtering in fastp (Chen et al., 2018) (v0.20.0) with the following parameters: -q 30 -w 16. The clean reads were taxonomically classified using Kaiju (Menzel et al., 2016) (v1.9.0) with the preformatted NR database (Mar. 10th, 2022).

5.5. ARG and MGE sequence identification and abundance calculation

In order to identify ARG and MGE sequences, we have developed a novel pipeline called ARGs-MGEs-MAP (AMM). The procedure for constructing the ARG database and the re-evaluation of cutoff parameters in the annotation pipeline are detailed in supplementary text. According to the evaluation results, a fragment was considered ARG-like or MGE-like if its top alignment to a reference sequence exhibited an e -value $\leq 10^{-7}$, similarity of $\geq 80\%$ and a length of ≥ 25 amino acids. The quantification of ARGs or MGEs was normalized to cell numbers (ARG copies/cell or MGE copies/cell), which were estimated using single-copy gene families (Nayfach and Pollard, 2015). Furthermore, the taxonomic classification of ARG or MGE sequences adhered to other described methodologies.

5.6. Metagenomic assembly and binning

The metagenomic sequences were assembled using MEGAHIT (Li et al., 2015) (v1.2.9) with the specified options: -t 22 -min-contig-len 1000. Subsequent refinement and reassembly of the bins were conducted using MetaWRAP (Uritskiy et al., 2018) (v1.2.3) processing modules. The initial extraction was performed utilizing MaxBin2 (Wu et al., 2014) (v2.2.5), MetaBAT2 (Kang et al., 2019) (v2.12.1), and CONCOCT (Alneberg et al., 2014) (v1.1.0). Evaluation of bin completeness and contamination was carried out using CheckM (Parks et al., 2015) (v1.1.3). Genomes meeting the criteria of containing less than 10 % contamination and more than 50 % genome completeness were deemed of substantial quality. The assembled genomes were then categorized into three levels based on their contamination and completeness: high (contamination $< 5\%$ and completeness $> 90\%$), medium (contamination $< 10\%$ and completeness $> 70\%$), and low (contamination $< 10\%$ and completeness $> 50\%$). Taxonomic classification of the generated MAGs was performed using GTDB-Tk (Chaumeil et al., 2020) (v2.2.0) and the 'gtdbtk classify_wf' command with the GTDB Release 207 database.

5.7. Gene prediction and annotation

The identification of open reading frames (ORFs) was accomplished using Prokka (Seemann, 2014) (v1.14.5). To uncover potential antibiotic resistance genes (ARGs) or mobile genetic elements (MGEs), the protein sequences derived from the ORFs were subjected to a BLASTP search against the reference database, employing a significance threshold of E -value 10^{-10} , as well as a minimum similarity and query coverage of 70 % (Hu et al., 2013). Virulence factors were annotated utilizing VFDB (Liu et al., 2022) set A (as of 01- January 2023) as the reference database. The antibiotics and secondary metabolites analysis shell (antiSMASH, v6.1.1) (Blin et al., 2021) tool was employed for the identification, annotation, and analysis of secondary metabolite biosynthesis gene clusters.

To identify potential transferable ARGs and the contributing MGEs, we identified on the co-occurrence of ARGs and MGEs within the same contig. To mitigate the impact of large contigs formed during the assembly process, we excluded instances where ARGs and MGEs were separated by 20 or more genes.

5.8. Phylogenetic tree

In order to reconstruct the phylogenetic structure of high-level genomes, a phylogenetic tree was constructed using PhyloPhlAn (Asnicar et al., 2020) (v3.0.3). The parameters were set as follows: "-db_aa

diamond -map_dna diamond -msa muscle -trim trimal -tree1 iqtree -tree2 raxml". Given the significant variations in genome contamination and completeness among carriers of antibiotic resistance genes (ARGs), these genomic distances were clustered into trees using mashree (Katz et al., 2019) (v1.2.0). To align with the sequences of specific ARG, multiple sequence alignments were generated using MAFFT (Katoh and Standley, 2013) (Multiple Alignment using Fast Fourier Transform, v7.490), and gapped regions were subsequently removed using trimAl (Capella-Gutiérrez et al., 2009) (v1.41). The phylogenetic trees were inferred using IQ-Tree (Nguyen et al., 2015) (1.6.12) with the following parameters: -b 1000 -alrt 1000 -m MF+MERGE. All the phylogenetic trees were visualized using ggtree (Yu et al., 2017) (v3.6.2) in the R package (v4.2.1).

5.9. Statistics and visualization

All statistical analyses were performed using R4.2.1 (<https://www.r-project.org/>). To evaluate the disparities in detection capabilities between the two ARG databases, we performed the paired-sample Wilcoxon signed-rank test on 12 samples. We analyzed the correlation between the total length of sequencing reads and the number of assembled contigs using the Pearson correlation coefficient, with significance determined through the "cor.test" function. The principal coordinate analysis (PCoA) analysis, Permutational multivariate analysis, and Procrustes analysis were analyzed using the "vegan" package based on Bray-Curtis distance. The quantitative relationship between collected reads of ARGs and MGEs was analyzed. Linear was selected to fit the two variables, aiming to determine the coefficient of determination (R^2). The potential relationships between bacterial communities, ARG subtypes and MGE were investigated using network analysis, which was implemented with the "vegan", "igraph" and "hmisc" packages and visualized using Gephi (Bastian et al., 2009) (v 0.9.2). A correlation between two nodes was considered statistically significant for Spearman's correlation coefficient $r \geq 0.7$ and the p -value ≤ 0.05 . The Bayesian-based SourceTracker (Knights et al., 2011) method was performed to quantify the extent of contribution of a potential source to the sink. We applied SourceTracker (Knights et al., 2011) (v1.0.1) to predict the degree to which ARGs were shared between rivers and sediment samples and then determine the source of the ARGs. The Sankey diagrams were generated using the "ggalluvial" package. For NMDS analysis, we used Bray-Curtis distances to assign genotypes (ARGs and MGEs). Moreover, we repeated the PAM clustering with Euclidean and Manhattan distances to test the consistency of the predicted clusters. To compare the medians of ARGs or MGEs, we employed the Mann-Whitney U test.

Maptree was utilized to characterize the composition and distribution of ARGs in sediment and water samples, based on the use of "ggraph", "igraph", and "tidyverse". The heatmaps were visualized using the "Pheatmap" package. The gene structure diagrams for different bins are generated using the "gggenes" package. Other plots were generated with the "ggplot2" package.

GeoDetector is a statistical tool designed for detecting impact mechanisms and identifying risk hotspots (Wang et al., 2010). It can discern the driving forces of multiple independent variables (x_i) on the dependent variable (y) as well as their interactions, without necessitating linear assumptions. Utilize the 'optdisc' function within the 'GD' package to transform continuous variables into categorical variables (Song et al., 2020). We employ a GeoDetector model to assess the impacts of eight explanatory variables (geographical location, reservoir characteristics, physical and chemical parameters, and quantitative methods) on ARGs and their interactions. The raw data and classification information pertaining to these variables can be found in Table S6. Among the 8 independent variables, those identified as having a significant effect (q -value: $p < 0.05$) by the factor detector of the GeoDetector model were further analyzed to assess the driving patterns of differences in the distribution of ARGs. Linear were chosen to fit two-variable on the R^2 (Burnham, 2002). The Analysis of Similarities

(ANOSIM) was utilized to assess whether there are disparities in ARG distribution caused by environmental media types (water vs sediment) with “vegan” package in R software.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve readability. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Sai Wang: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Formal analysis, Data curation. **Wenhan Nie:** Writing – review & editing, Writing – original draft, Resources, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qing Gu:** Resources, Funding acquisition. **Xie Wang:** Resources, Methodology. **Danping Yang:** Resources. **Hongyu Li:** Resources. **Peihong Wang:** Resources, Data curation. **Weixue Liao:** Resources. **Jin Huang:** Resources, Formal analysis. **Quan Yuan:** Resources. **Shengli Zhou:** Resources. **Iftikhar Ahmad:** Writing – review & editing. **Kiga Kotaro:** Writing – review & editing. **Gongyou Chen:** Writing – review & editing, Validation. **Bo Zhu:** Conceptualization, Funding acquisition, Investigation, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The sequencing data generated in this study is available and can be accessed through NCBI BioProject PRNJA919165. The enhanced curated database tool named ARGs-MGEs-MAP pipeline in this study can be found on Github at <https://www.github.com/bioone/ARGs-MGEs-MAP>. The 15,117 metagenomic assembled genomes are available at https://www.figshare.com/articles/dataset/metagenomic_assembly_genome/23936307.

Acknowledgments

We thank anonymous reviewers for their excellent comments that improved the manuscript. This manuscript was edited by Wallace Academic Editing. This work was supported by National Natural Science Foundation of China (32200142, 32272479), Open Project Program of State Key Laboratory of Rice Biology (20190109), Open Project Program of State Key Laboratory for Biology of Plant Diseases and Insect Pests (SKLOF202201), Zhejiang Province ecological Environment Research and Promotion Project (2020HT0009), Shanghai Committee of Science and Technology (19390743300 and 21ZR1435500), Chongqing Natural Science Foundation (CSTB2022NSCQ-MSX0524) and C-FINDs International Exchange Research Grant (2023-CF-02).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2024.121572](https://doi.org/10.1016/j.watres.2024.121572).

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